Pikachu Metrics 数据分析 metrics_Mjlab-Tracking-Flat-Pikachu-Walk-Base__model_4999_20260821_111145_with_target

FILE
metrics_Mjlab-Tracking-Flat-Pikachu-Walk-Base__model_4999_20260821_111145_with_target.csv

NETWORK
Mjlab-Tracking-Flat-Pikachu-Walk-Base__model_4999

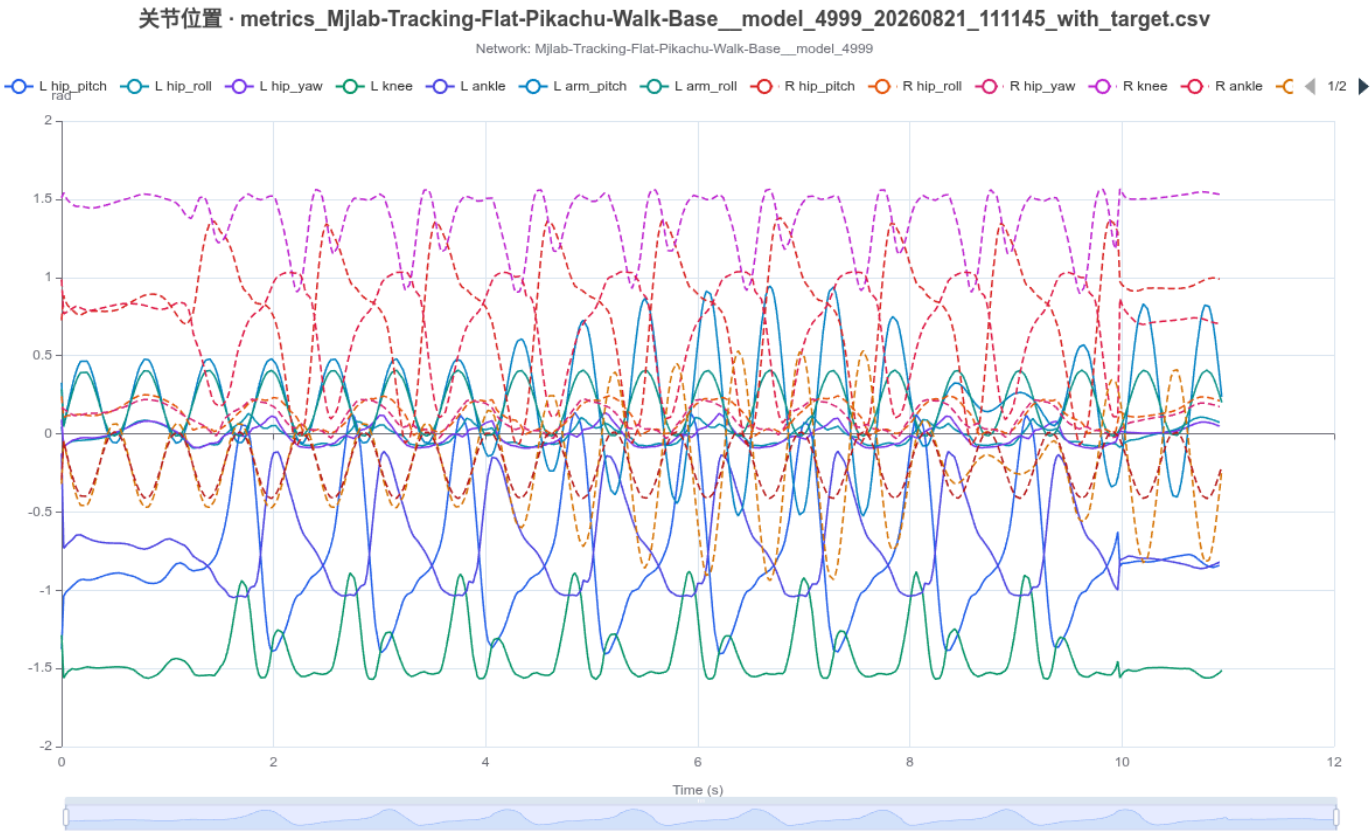
SAMPLES
1,409

DURATION
30.320 s

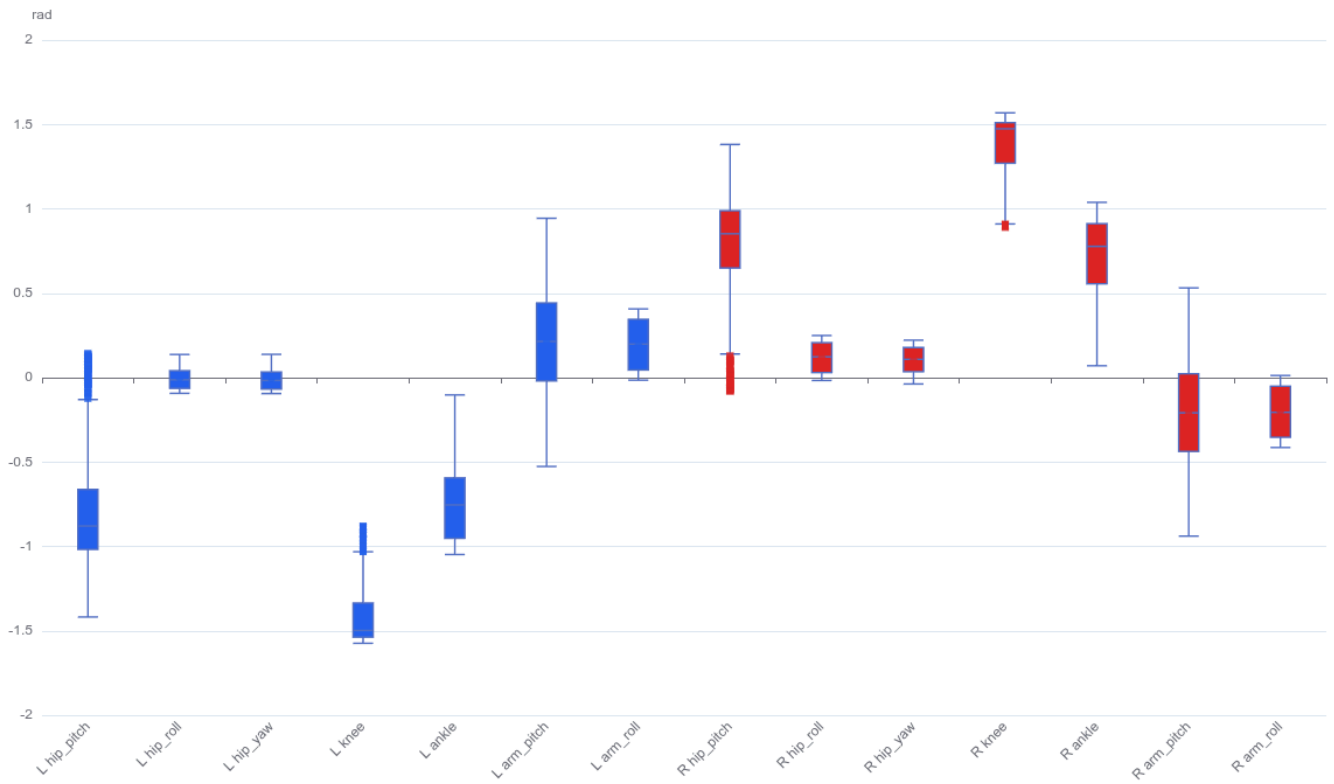
ENVIRONMENTS
1

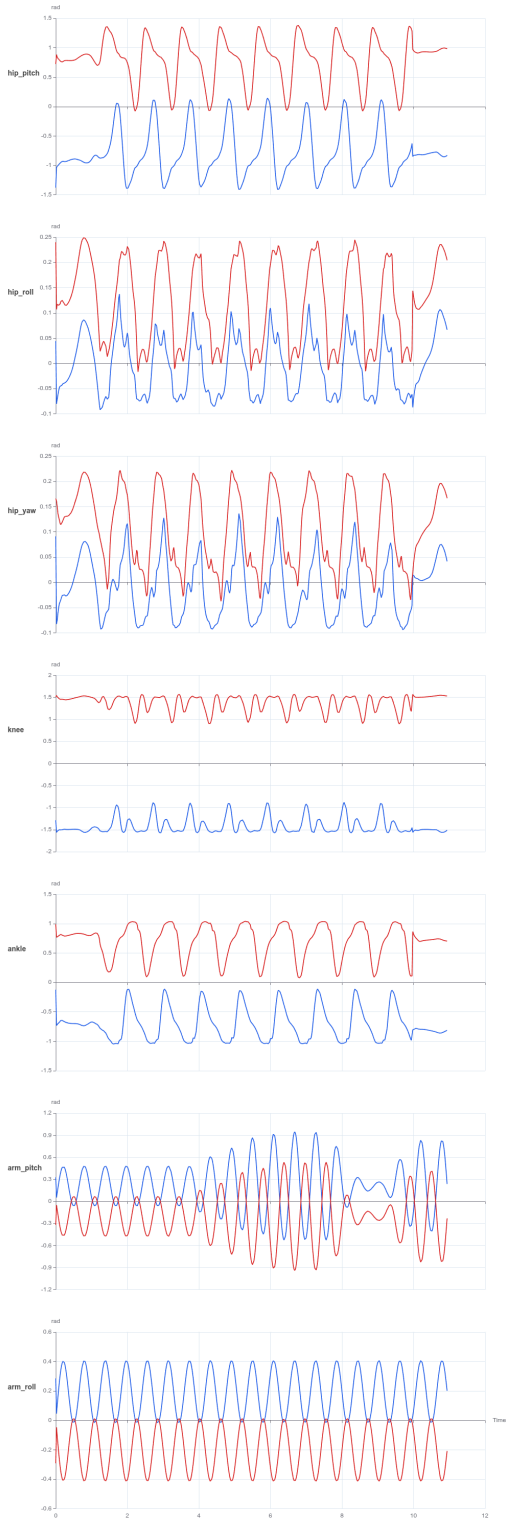
时间区间: 0.000 — 10.940 s · 仅含已勾选字段

关节位置

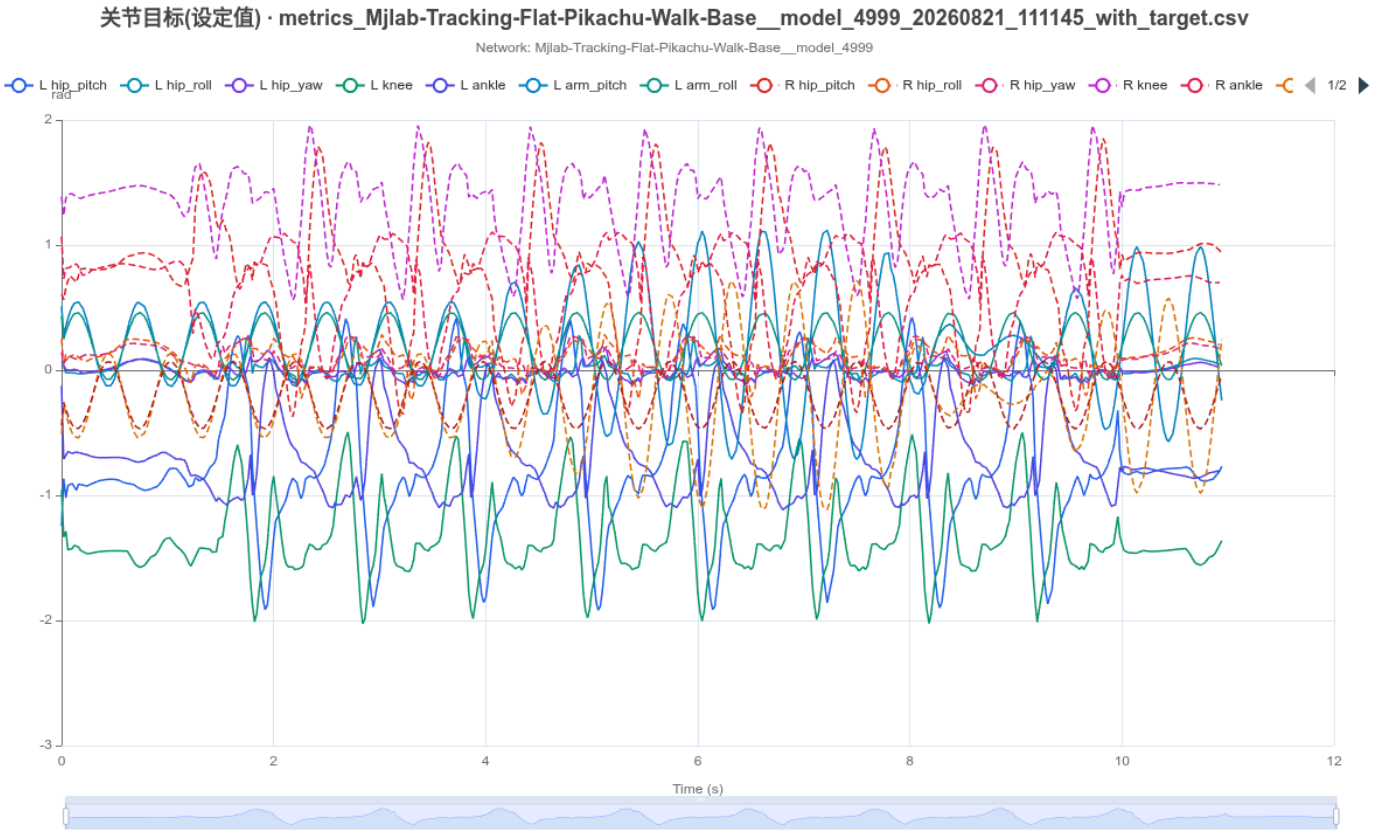


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	503	-1.41841	0.144098	-0.792156	0.889807	1.41841	1.36448
joint_position_rad/left_hip_roll_joint	10	15	503	-0.0931600	0.137381	-0.00687251	0.0583765	0.137381	0.0906361
joint_position_rad/left_hip_yaw_joint	10	15	503	-0.0943847	0.137774	-0.0119419	0.0608729	0.137774	0.0934061
joint_position_rad/left_knee_joint	30	10	503	-1.57408	-0.880036	-1.41344	1.42517	1.57408	1.56112
joint_position_rad/left_ankle_joint	10	15	503	-1.04789	-0.102592	-0.713110	0.763037	1.04789	1.03564
joint_position_rad/left_arm_pitch_joint	10	15	503	-0.526216	0.944287	0.213299	0.392400	0.944287	0.796598
joint_position_rad/left_arm_roll_joint	10	15	503	-0.0148722	0.407248	0.195520	0.246257	0.407248	0.403491
joint_position_rad/right_hip_pitch_joint	30	10	503	-0.0807430	1.38148	0.784624	0.875301	1.38148	1.33201
joint_position_rad/right_hip_roll_joint	10	15	503	-0.0166893	0.249369	0.121429	0.146850	0.249369	0.233510
joint_position_rad/right_hip_yaw_joint	10	15	503	-0.0375236	0.222159	0.106282	0.130330	0.222159	0.215107
joint_position_rad/right_knee_joint	30	10	503	0.889994	1.56863	1.38183	1.39373	1.56863	1.54619
joint_position_rad/right_ankle_joint	10	15	503	0.0703060	1.03868	0.704493	0.758977	1.03868	1.03057
joint_position_rad/right_arm_pitch_joint	10	15	503	-0.938988	0.531870	-0.207683	0.389455	0.938988	0.790945
joint_position_rad/right_arm_roll_joint	10	15	503	-0.414048	0.0122132	-0.201804	0.251878	0.414048	0.410291

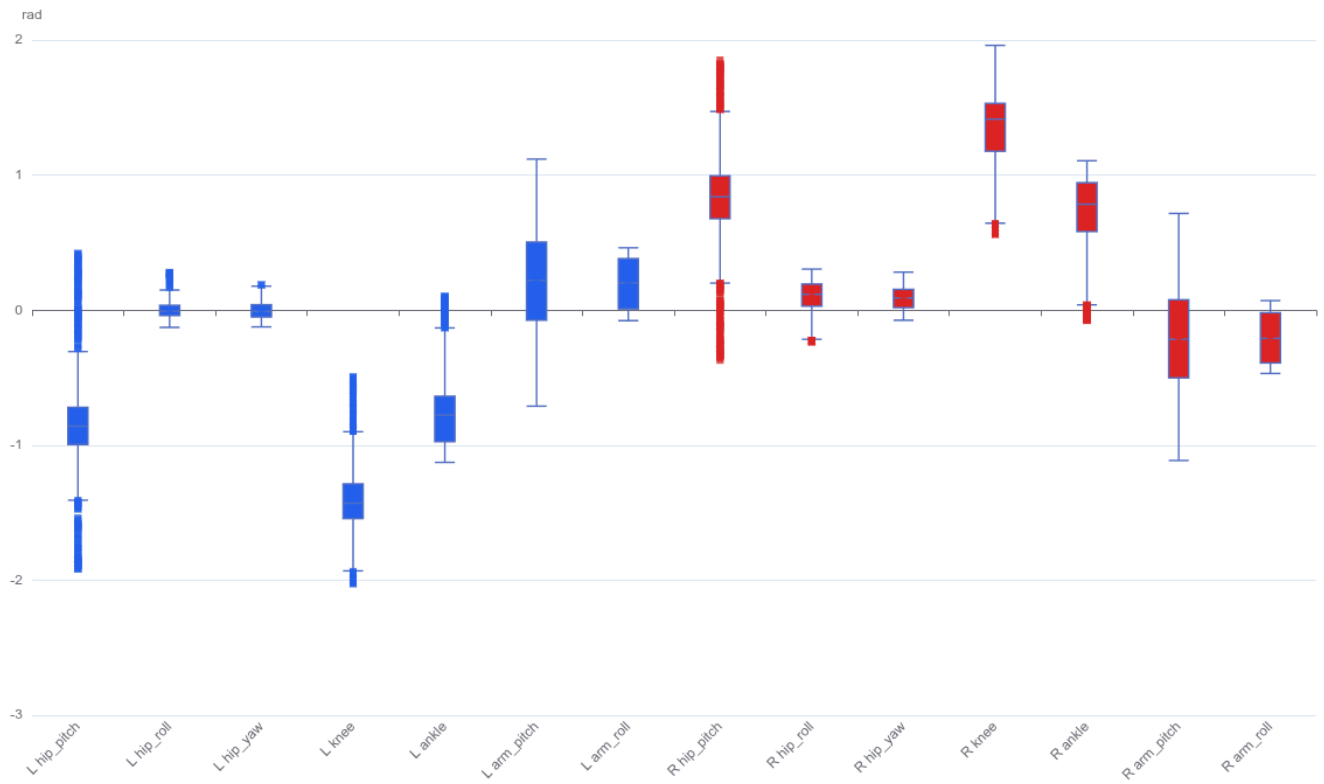

rad
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

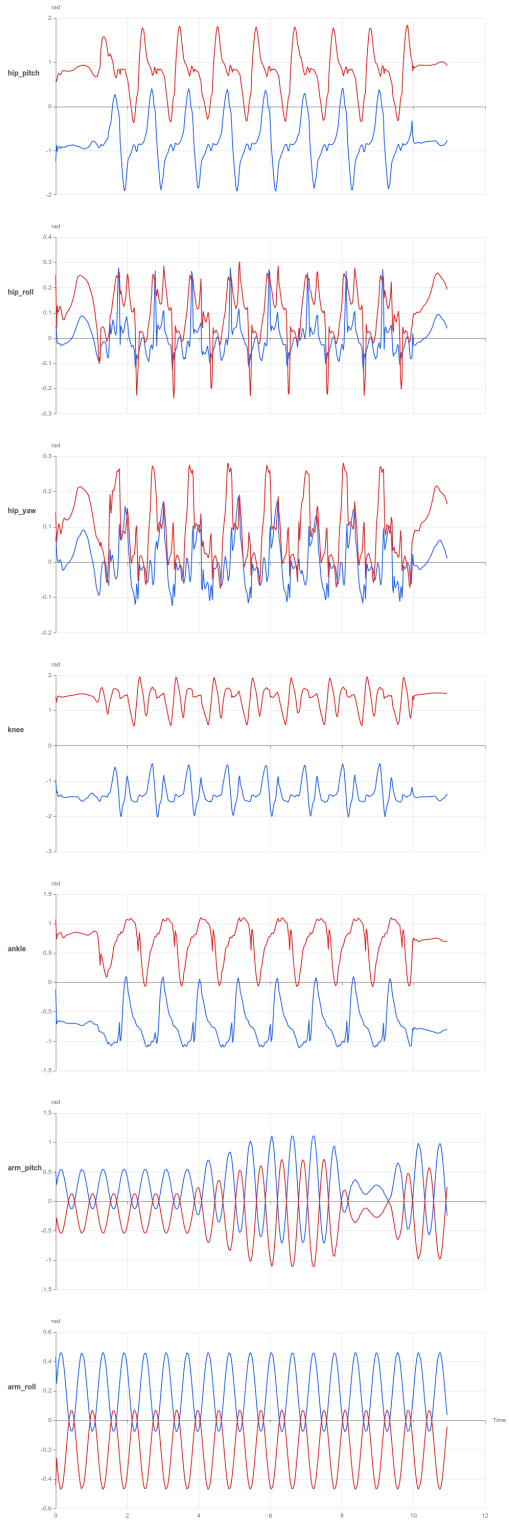

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节目标(设定值)

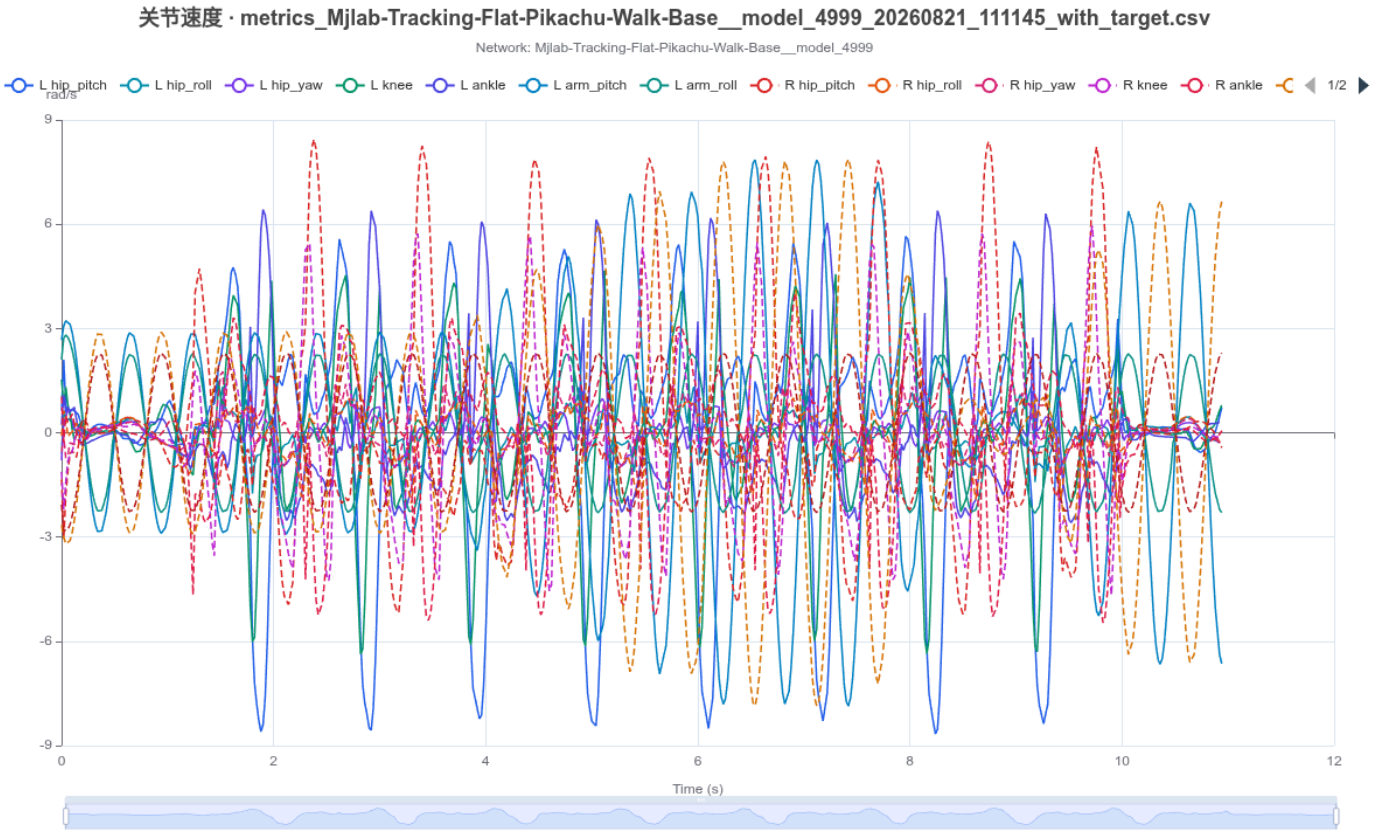


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_target_rad/left_hip_pitch_joint	30	10	503	-1.91746	0.420984	-0.799819	0.946405	1.91746	1.72845
joint_target_rad/left_hip_roll_joint	10	15	503	-0.128128	0.278996	0.00222641	0.0686229	0.278996	0.114252
joint_target_rad/left_hip_yaw_joint	10	15	503	-0.124393	0.187954	-0.00100444	0.0643719	0.187954	0.122428
joint_target_rad/left_knee_joint	30	10	503	-2.02581	-0.493546	-1.36057	1.39531	2.02581	1.78495
joint_target_rad/left_ankle_joint	10	15	503	-1.12785	0.102380	-0.719634	0.782597	1.12785	1.08349
joint_target_rad/left_arm_pitch_joint	10	15	503	-0.711751	1.11781	0.214078	0.464665	1.11781	0.929965
joint_target_rad/left_arm_roll_joint	10	15	503	-0.0782836	0.461716	0.195309	0.272503	0.461716	0.458122
joint_target_rad/right_hip_pitch_joint	30	10	503	-0.371215	1.85237	0.794816	0.937356	1.85237	1.68776
joint_target_rad/right_hip_roll_joint	10	15	503	-0.237724	0.303844	0.105953	0.151760	0.303844	0.250088
joint_target_rad/right_hip_yaw_joint	10	15	503	-0.0749552	0.281279	0.0935879	0.129309	0.281279	0.259718
joint_target_rad/right_knee_joint	30	10	503	0.559567	1.96046	1.34256	1.37807	1.96046	1.80616
joint_target_rad/right_ankle_joint	10	15	503	-0.0773152	1.10596	0.702202	0.773242	1.10596	1.07968
joint_target_rad/right_arm_pitch_joint	10	15	503	-1.11360	0.715964	-0.209864	0.462739	1.11360	0.925752
joint_target_rad/right_arm_roll_joint	10	15	503	-0.469196	0.0708036	-0.202789	0.277913	0.469196	0.465602

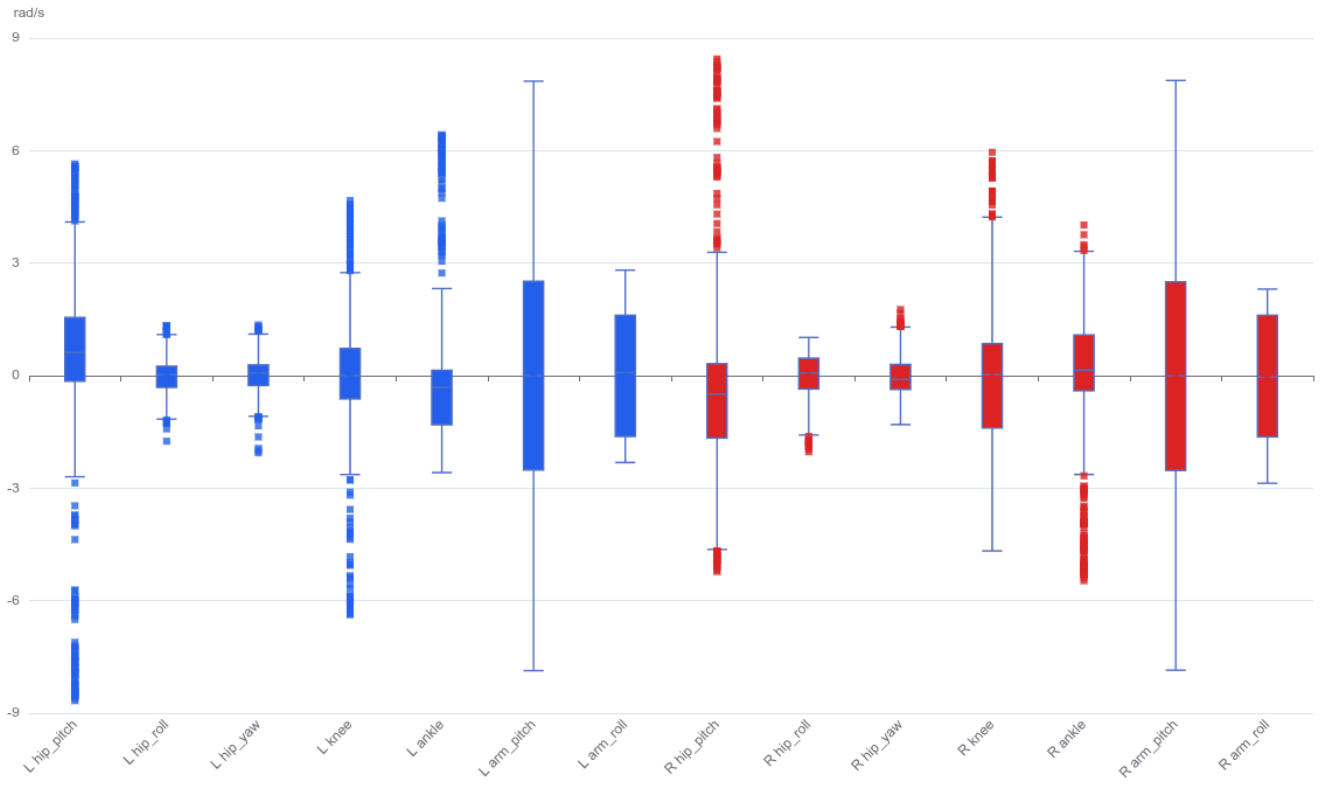

rad
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

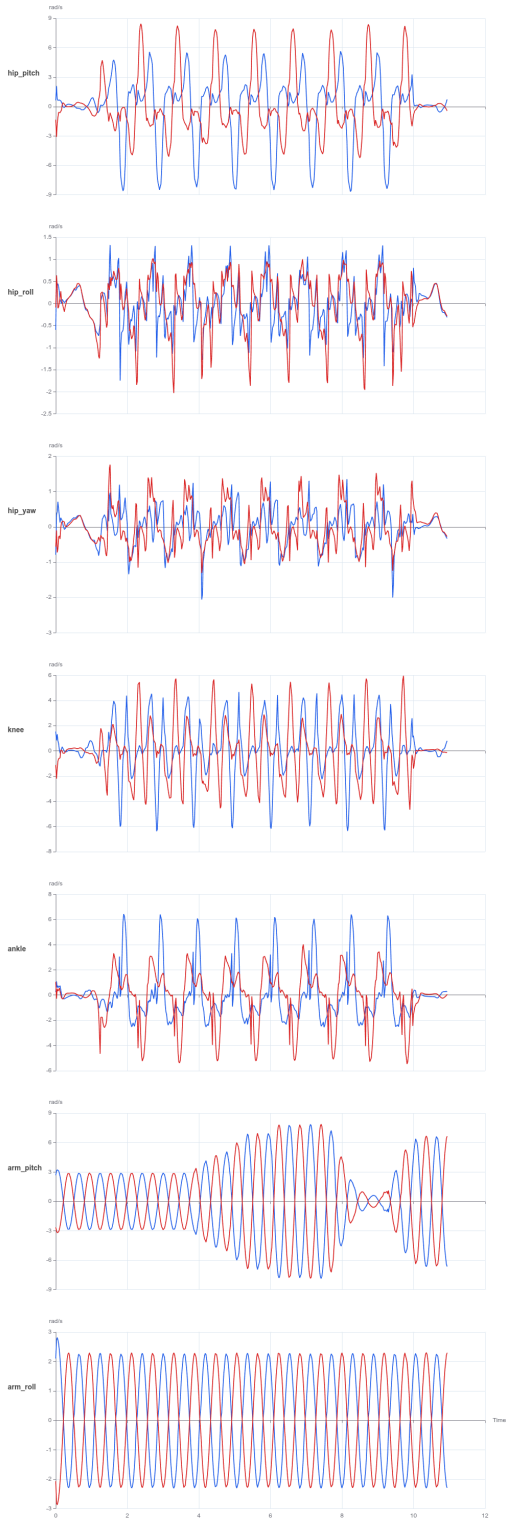

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度

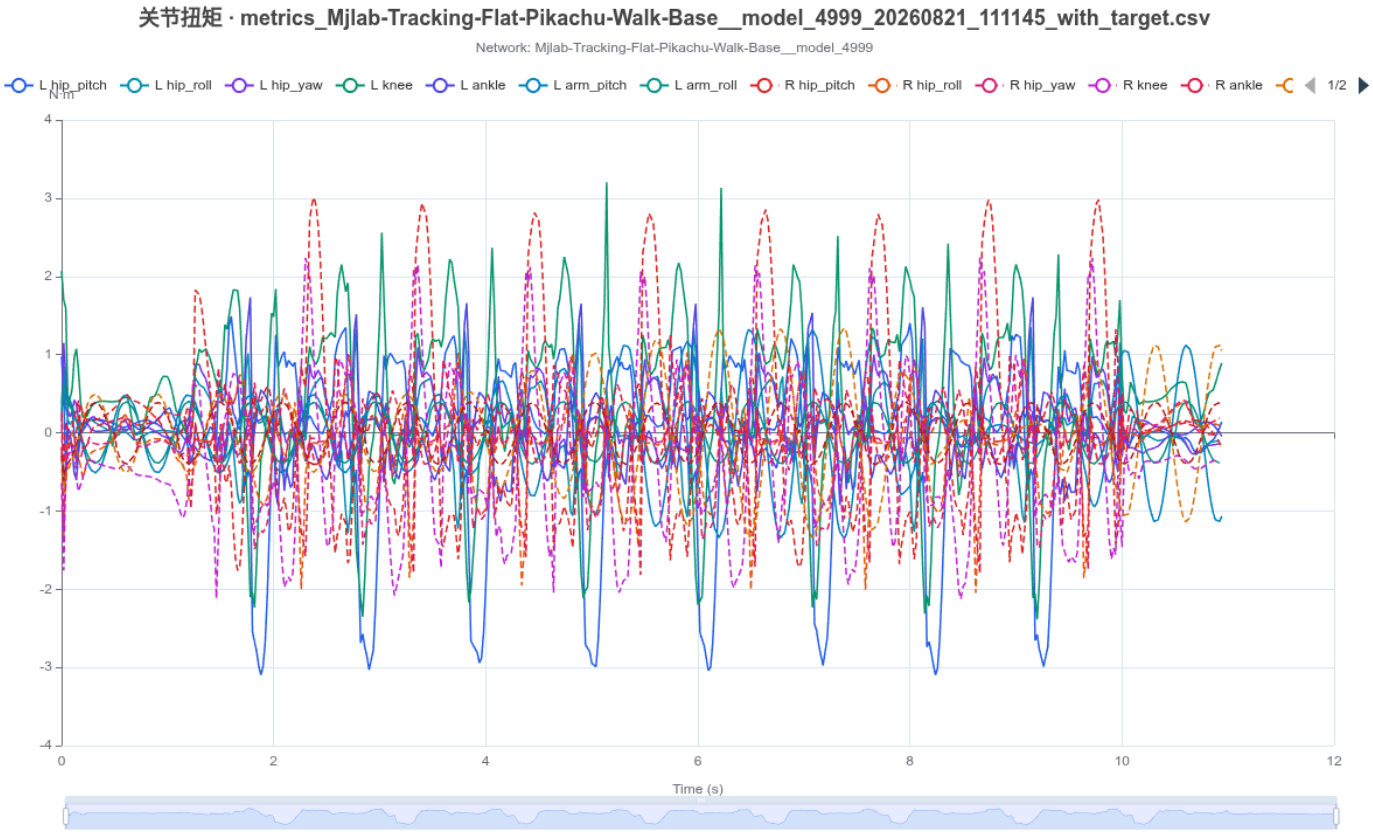


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	503	-8.66506	5.64462	0.0468706	3.22467	8.66506	7.80209
joint_velocity_rad_s/left_hip_roll_joint	10	15	503	-1.74865	1.32564	-0.00142227	0.504410	1.74865	1.04003
joint_velocity_rad_s/left_hip_yaw_joint	10	15	503	-2.05752	1.35162	-0.00522079	0.502841	2.05752	0.962143
joint_velocity_rad_s/left_knee_joint	30	10	503	-6.37312	4.66915	0.00665887	2.14191	6.37312	4.56185
joint_velocity_rad_s/left_ankle_joint	10	15	503	-2.58458	6.42226	-0.0610370	2.02908	6.42226	5.41669
joint_velocity_rad_s/left_arm_pitch_joint	10	15	503	-7.86561	7.84891	0.0259673	3.53928	7.86561	6.95324
joint_velocity_rad_s/left_arm_roll_joint	10	15	503	-2.31653	2.81334	0.0176695	1.62234	2.81334	2.28186
joint_velocity_rad_s/right_hip_pitch_joint	30	10	503	-5.22137	8.44342	0.0512284	3.10301	8.44342	7.51174
joint_velocity_rad_s/right_hip_roll_joint	10	15	503	-2.02972	1.01935	-0.00191449	0.619524	2.02972	1.19604
joint_velocity_rad_s/right_hip_yaw_joint	10	15	503	-1.30395	1.76211	0.00119776	0.579081	1.76211	1.14153
joint_velocity_rad_s/right_knee_joint	30	10	503	-4.67319	5.95471	-0.0337211	2.18578	5.95471	4.55440
joint_velocity_rad_s/right_ankle_joint	10	15	503	-5.46820	4.01577	-0.0919104	2.03293	5.46820	4.61874
joint_velocity_rad_s/right_arm_pitch_joint	10	15	503	-7.85311	7.86797	-0.0261258	3.54001	7.86797	6.95458
joint_velocity_rad_s/right_arm_roll_joint	10	15	503	-2.87105	2.30533	-0.0187848	1.63277	2.87105	2.28750

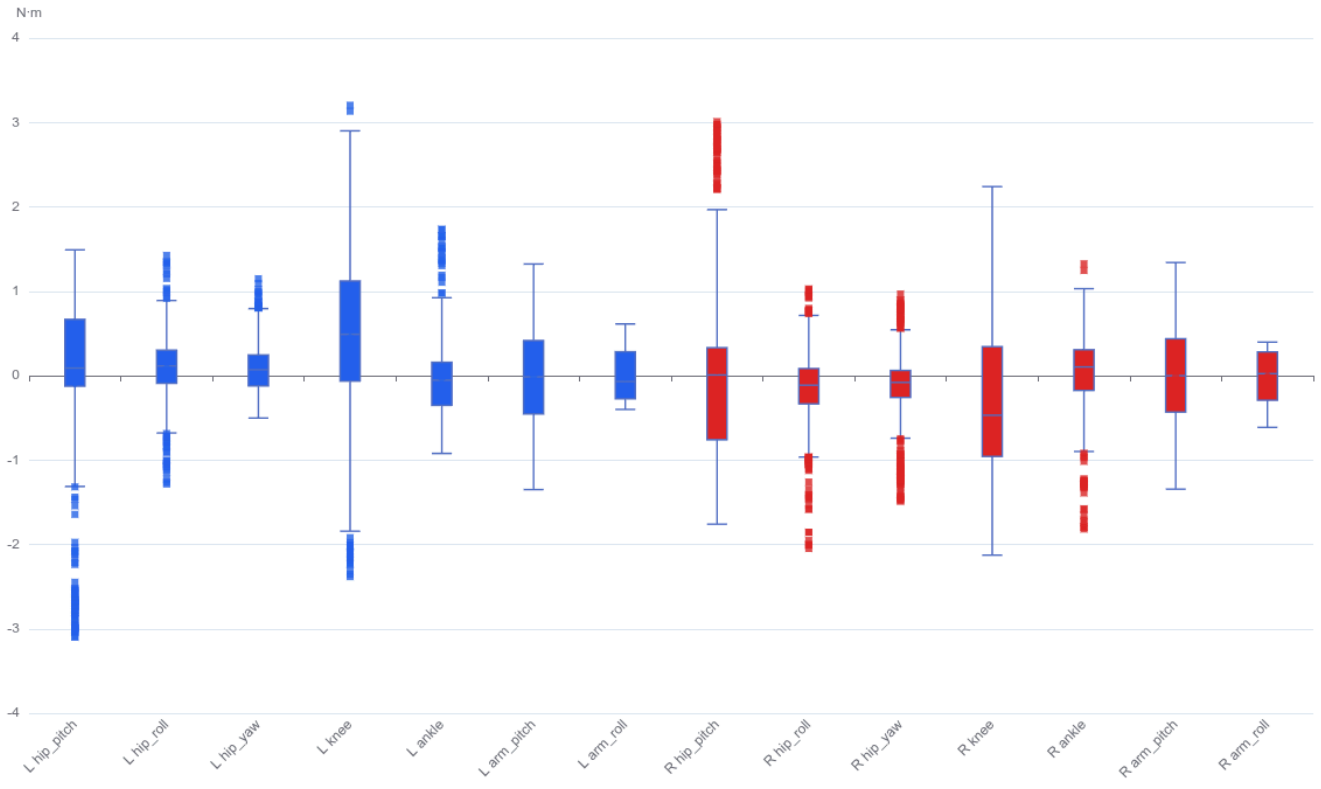

rad/s
9
6
3
0
-3
-6
-9
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

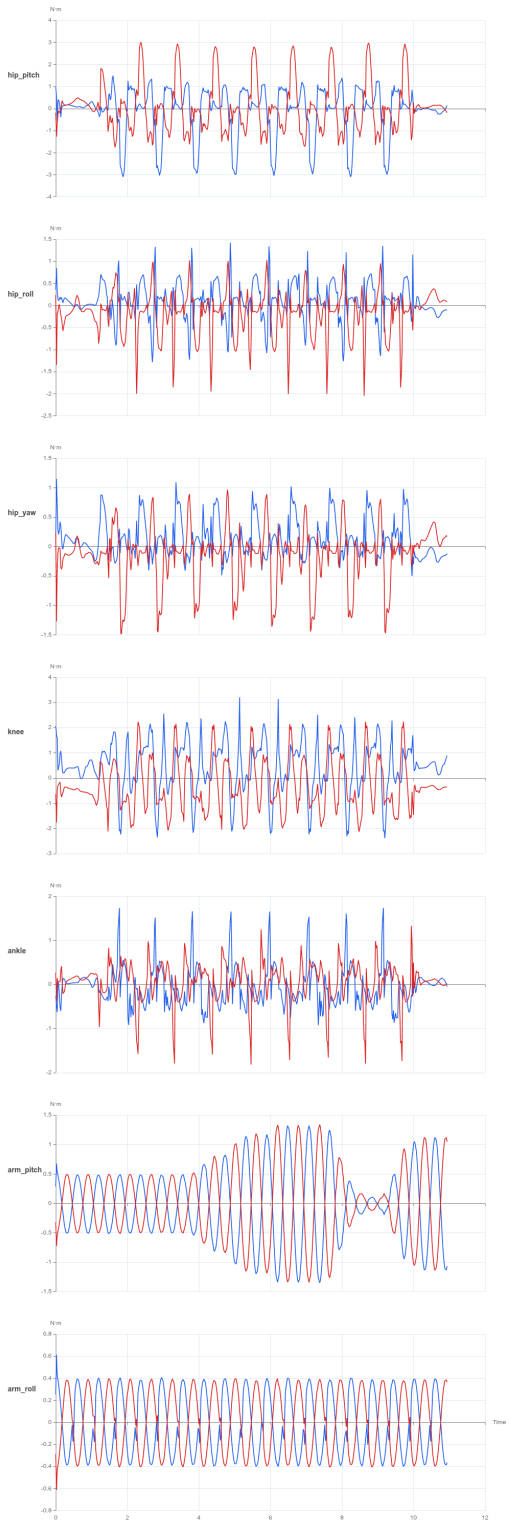

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad/s
Time

关节扭矩

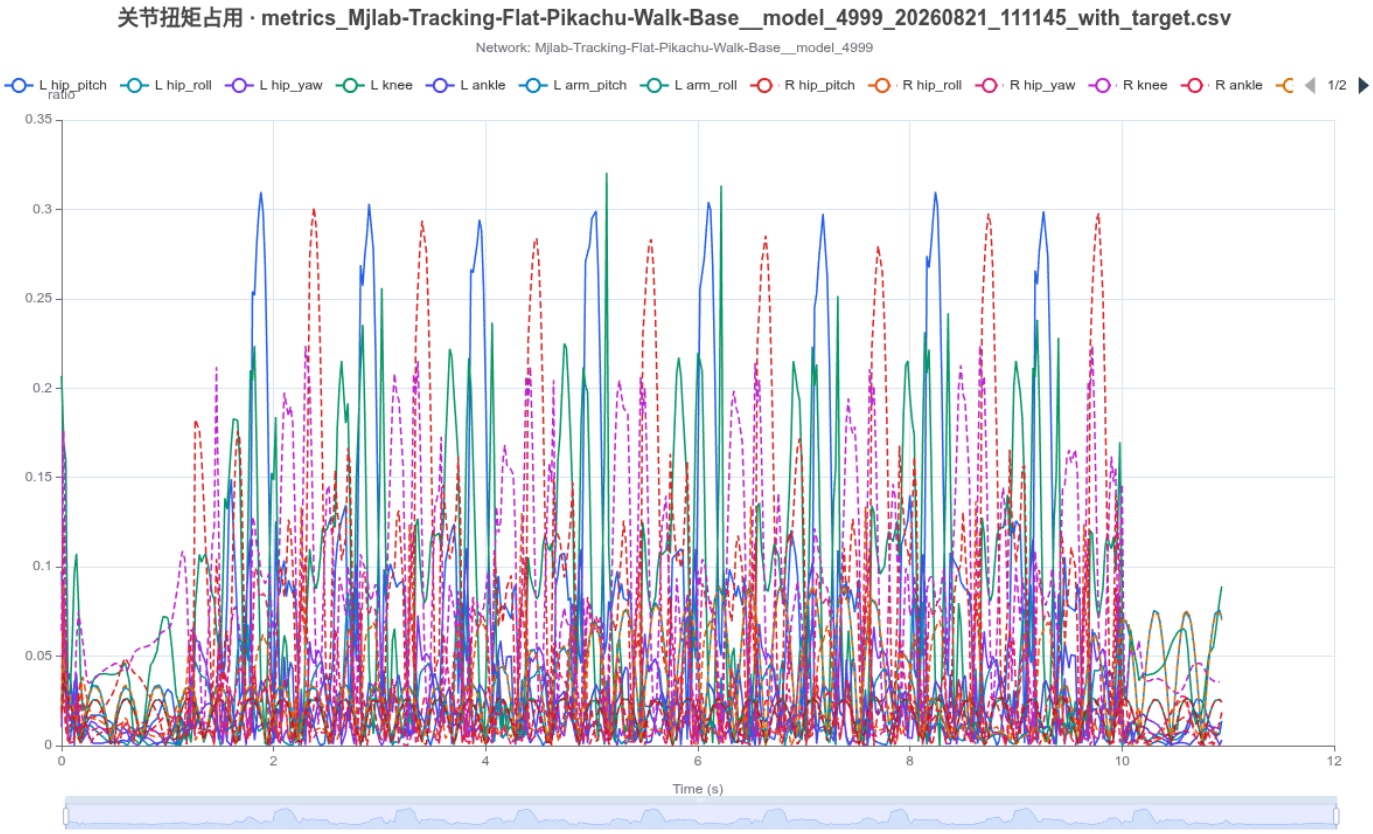


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	503	-3.09713	1.49015	-0.0930630	1.09602	3.09713	2.74391
joint_torque_nm/left_hip_roll_joint	10	15	503	-1.28459	1.42495	0.0994001	0.426462	1.42495	0.910512
joint_torque_nm/left_hip_yaw_joint	10	15	503	-0.502629	1.14822	0.128210	0.346808	1.14822	0.768549
joint_torque_nm/left_knee_joint	30	10	503	-2.38230	3.20420	0.457767	1.09974	3.20420	2.12634
joint_torque_nm/left_ankle_joint	10	15	503	-0.922183	1.73371	-0.0379742	0.452416	1.73371	0.792410
joint_torque_nm/left_arm_pitch_joint	10	15	503	-1.35082	1.32242	-0.0118810	0.608868	1.35082	1.17211
joint_torque_nm/left_arm_roll_joint	10	15	503	-0.400449	0.612586	-0.0156502	0.282060	0.612586	0.395425
joint_torque_nm/right_hip_pitch_joint	30	10	503	-1.75990	3.01063	0.0613130	1.12177	3.01063	2.68271
joint_torque_nm/right_hip_roll_joint	10	15	503	-2.04838	1.02772	-0.174759	0.533214	2.04838	1.03544
joint_torque_nm/right_hip_yaw_joint	10	15	503	-1.48940	0.967554	-0.146176	0.487009	1.48940	1.17872
joint_torque_nm/right_knee_joint	30	10	503	-2.12710	2.23971	-0.323889	1.03722	2.23971	1.97433
joint_torque_nm/right_ankle_joint	10	15	503	-1.81774	1.32526	0.0389706	0.466544	1.81774	0.980137
joint_torque_nm/right_arm_pitch_joint	10	15	503	-1.34507	1.34004	-0.00470215	0.609947	1.34507	1.17547
joint_torque_nm/right_arm_roll_joint	10	15	503	-0.612973	0.397143	0.00257916	0.281674	0.612973	0.396046

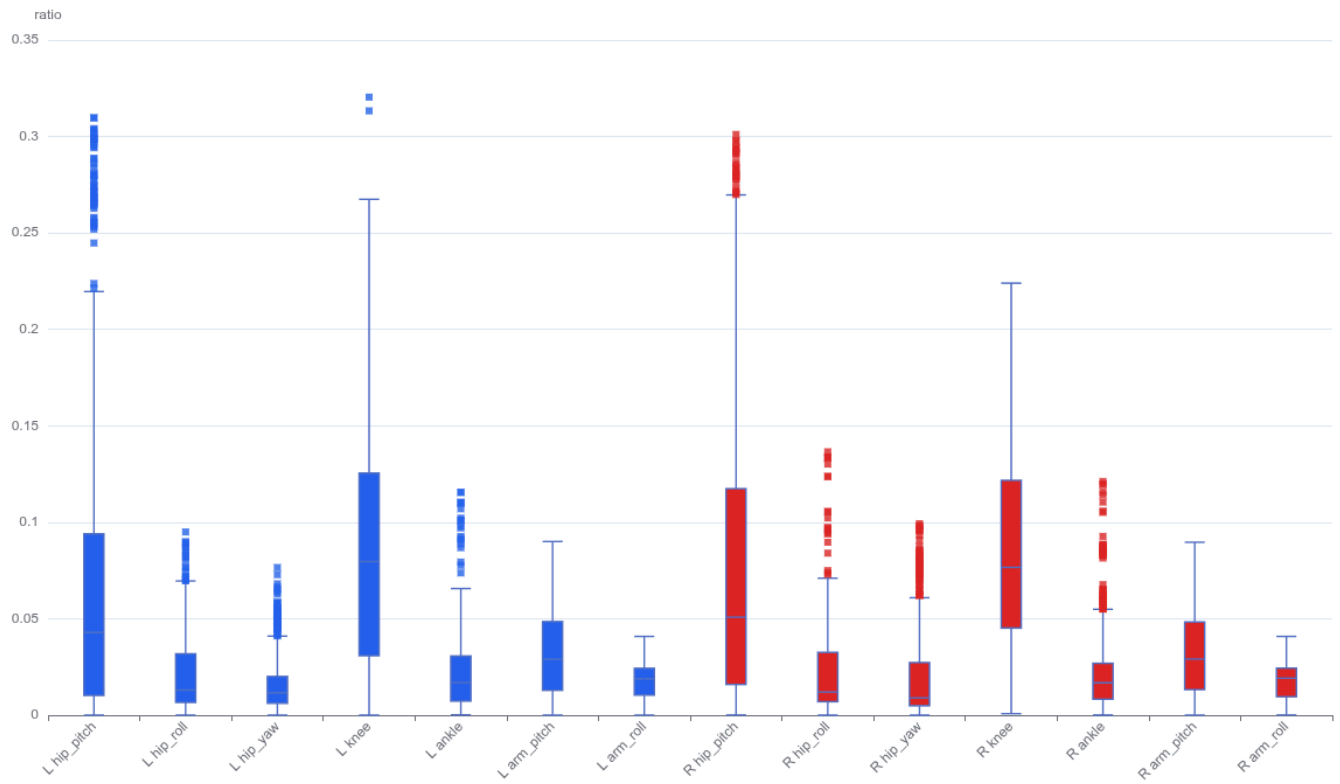

N·m
4
3
2
1
0
-1
-2
-3
-4
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

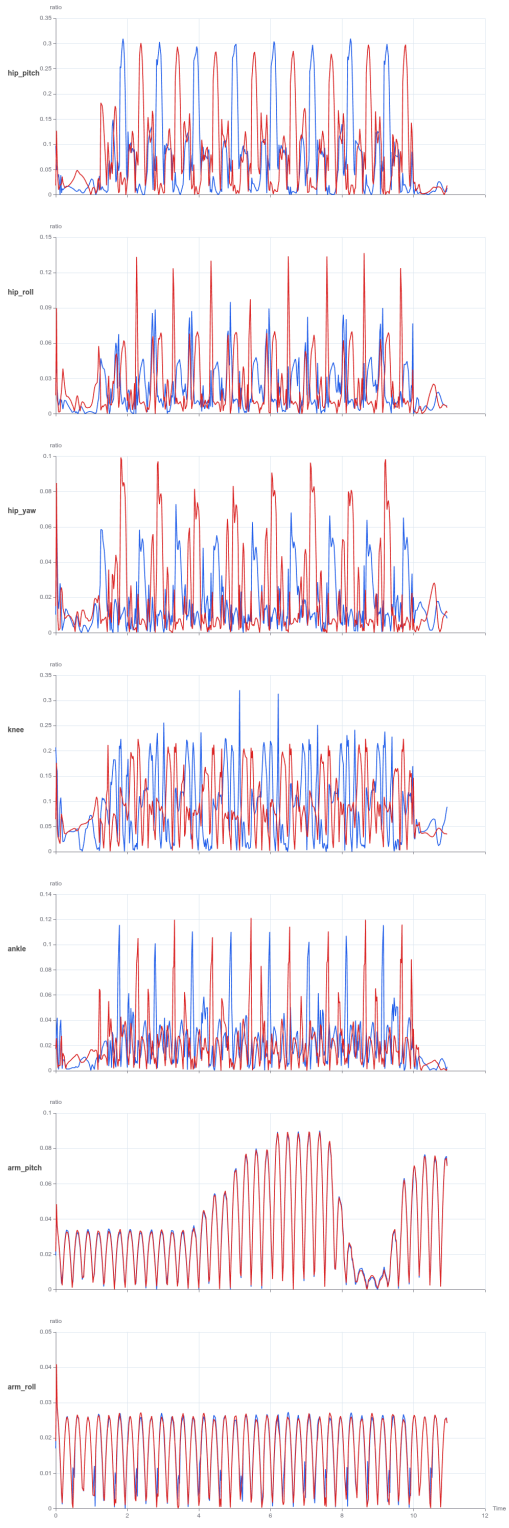

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

关节扭矩占用

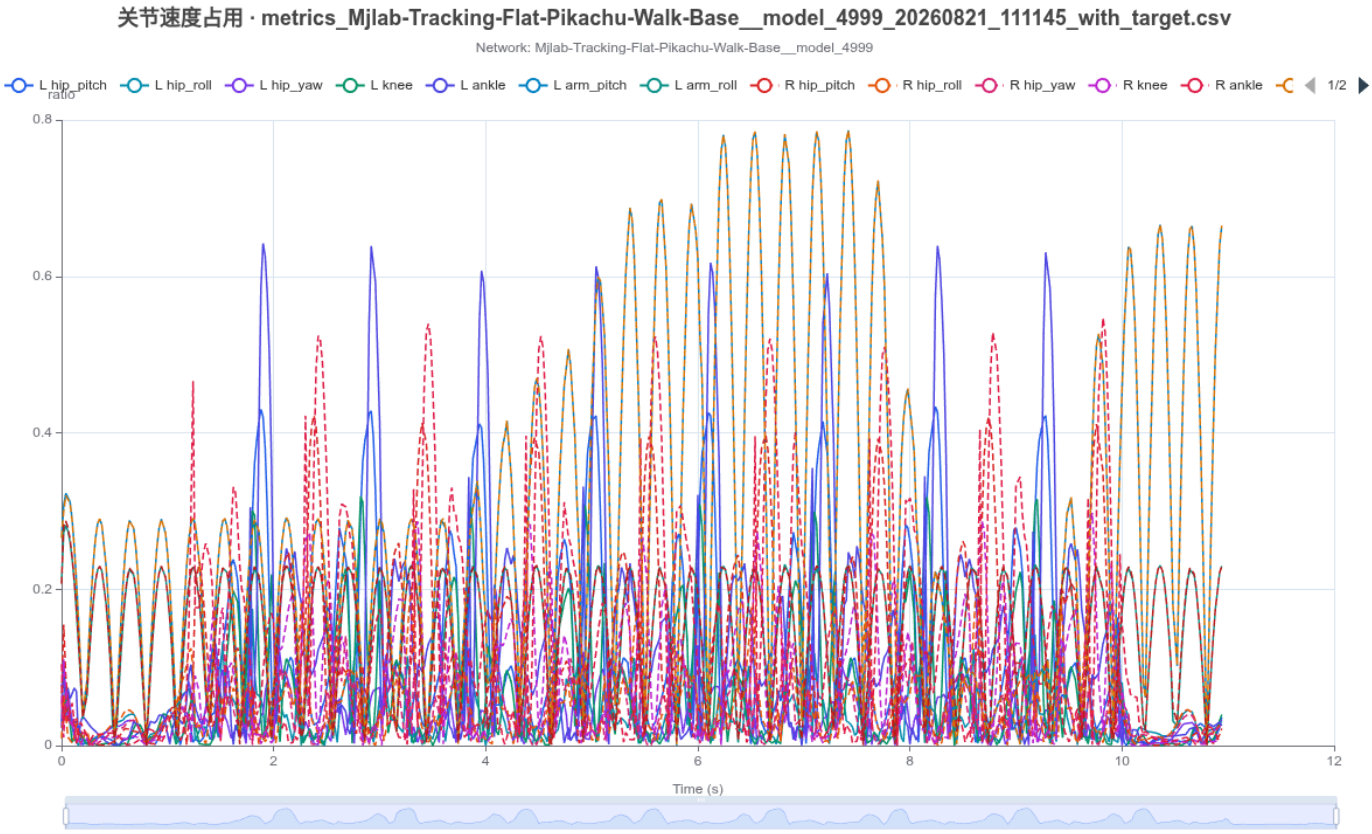


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	503	0.0000344007	0.309713	0.0716170	0.109602	0.309713	0.274391
joint_torque_ratio/left_hip_roll_joint	10	15	503	0.0000155508	0.0949966	0.0206944	0.0284308	0.0949966	0.0607008
joint_torque_ratio/left_hip_yaw_joint	10	15	503	0.0000149627	0.0765479	0.0169230	0.0231205	0.0765479	0.0512366
joint_torque_ratio/left_knee_joint	30	10	503	0.00000560679	0.320420	0.0874624	0.109974	0.320420	0.212634
joint_torque_ratio/left_ankle_joint	10	15	503	0.000174741	0.115580	0.0219972	0.0301611	0.115580	0.0528273
joint_torque_ratio/left_arm_pitch_joint	10	15	503	0.0000549381	0.0900545	0.0329200	0.0405912	0.0900545	0.0781404
joint_torque_ratio/left_arm_roll_joint	10	15	503	0.0000239615	0.0408390	0.0169534	0.0188040	0.0408390	0.0263616
joint_torque_ratio/right_hip_pitch_joint	30	10	503	0.0000830992	0.301063	0.0792784	0.112177	0.301063	0.268271
joint_torque_ratio/right_hip_roll_joint	10	15	503	0.00000231404	0.136559	0.0242481	0.0355476	0.136559	0.0690294
joint_torque_ratio/right_hip_yaw_joint	10	15	503	0.00000353282	0.0992936	0.0210967	0.0324672	0.0992936	0.0785816
joint_torque_ratio/right_knee_joint	30	10	503	0.000822494	0.223971	0.0880833	0.103722	0.223971	0.197433
joint_torque_ratio/right_ankle_joint	10	15	503	0.0000799181	0.121183	0.0222900	0.0311030	0.121183	0.0653424
joint_torque_ratio/right_arm_pitch_joint	10	15	503	0.0000559411	0.0896710	0.0329693	0.0406631	0.0896710	0.0783650
joint_torque_ratio/right_arm_roll_joint	10	15	503	0.000108932	0.0408649	0.0166906	0.0187782	0.0408649	0.0264031

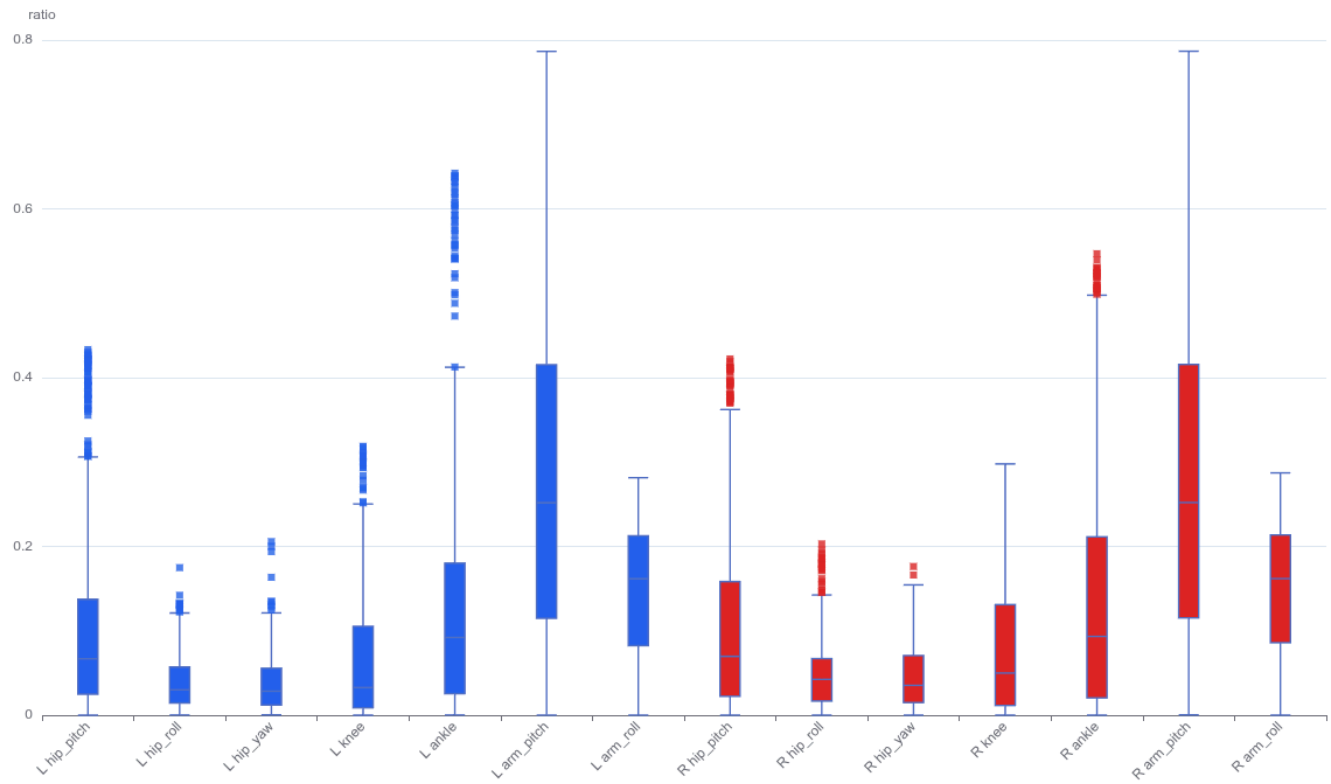

ratio
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

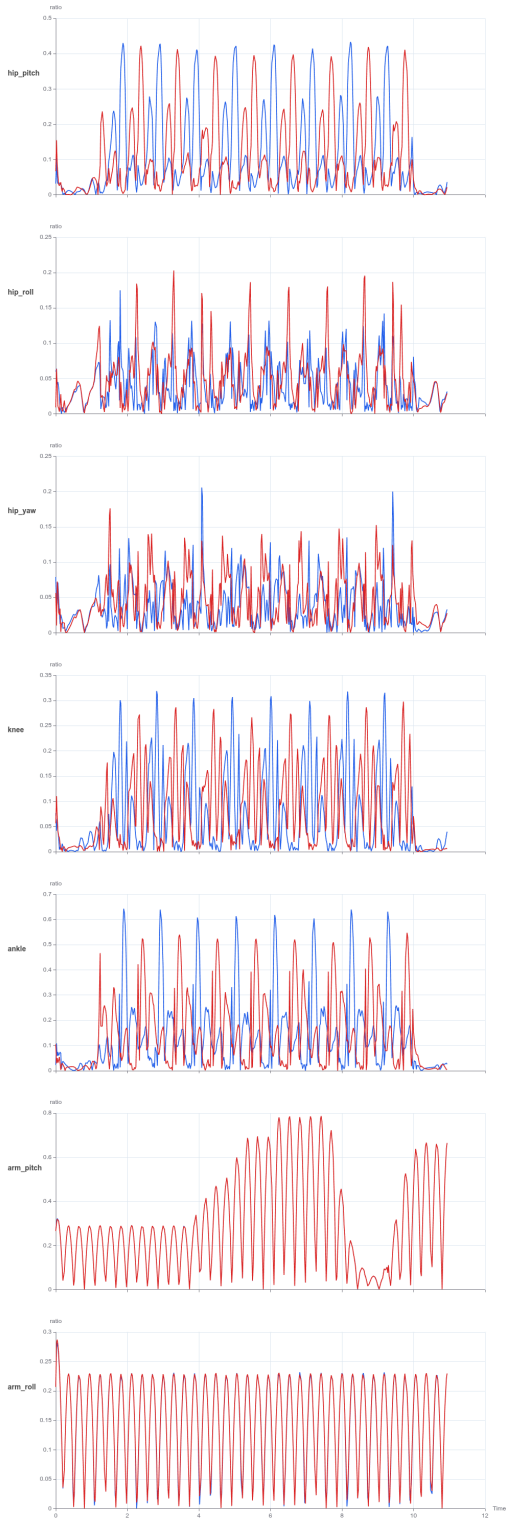

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节速度占用

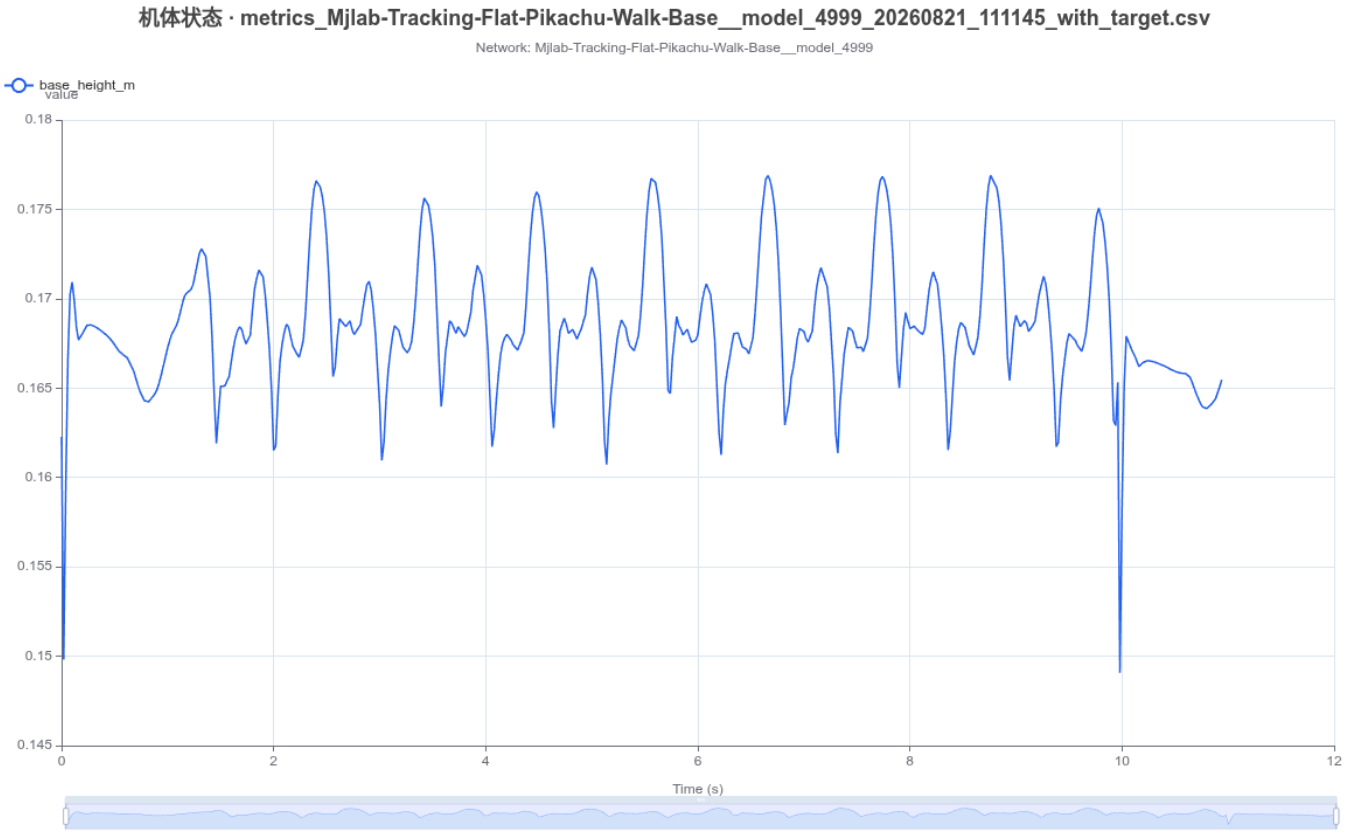


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	503	7.07476e-7	0.433253	0.109461	0.161234	0.433253	0.390105
joint_velocity_ratio/left_hip_roll_joint	10	15	503	0.000214205	0.174865	0.0390186	0.0504410	0.174865	0.104003
joint_velocity_ratio/left_hip_yaw_joint	10	15	503	0.000331235	0.205752	0.0377961	0.0502841	0.205752	0.0962143
joint_velocity_ratio/left_knee_joint	30	10	503	0.00000471446	0.318656	0.0702091	0.107095	0.318656	0.228092
joint_velocity_ratio/left_ankle_joint	10	15	503	0.000238147	0.642226	0.136779	0.202908	0.642226	0.541669
joint_velocity_ratio/left_arm_pitch_joint	10	15	503	0.000133007	0.786561	0.286085	0.353928	0.786561	0.695324
joint_velocity_ratio/left_arm_roll_joint	10	15	503	0.000125892	0.281334	0.144571	0.162234	0.281334	0.228186
joint_velocity_ratio/right_hip_pitch_joint	30	10	503	0.0000631032	0.422171	0.107825	0.155150	0.422171	0.375587
joint_velocity_ratio/right_hip_roll_joint	10	15	503	0.0000793212	0.202972	0.0481915	0.0619524	0.202972	0.119604
joint_velocity_ratio/right_hip_yaw_joint	10	15	503	0.0000999483	0.176211	0.0453656	0.0579081	0.176211	0.114153
joint_velocity_ratio/right_knee_joint	30	10	503	0.0000859254	0.297735	0.0785664	0.109289	0.297735	0.227720
joint_velocity_ratio/right_ankle_joint	10	15	503	0.0000348913	0.546820	0.142188	0.203293	0.546820	0.461874
joint_velocity_ratio/right_arm_pitch_joint	10	15	503	0.000347180	0.786797	0.286096	0.354001	0.786797	0.695458
joint_velocity_ratio/right_arm_roll_joint	10	15	503	0.000114675	0.287105	0.146152	0.163277	0.287105	0.228750

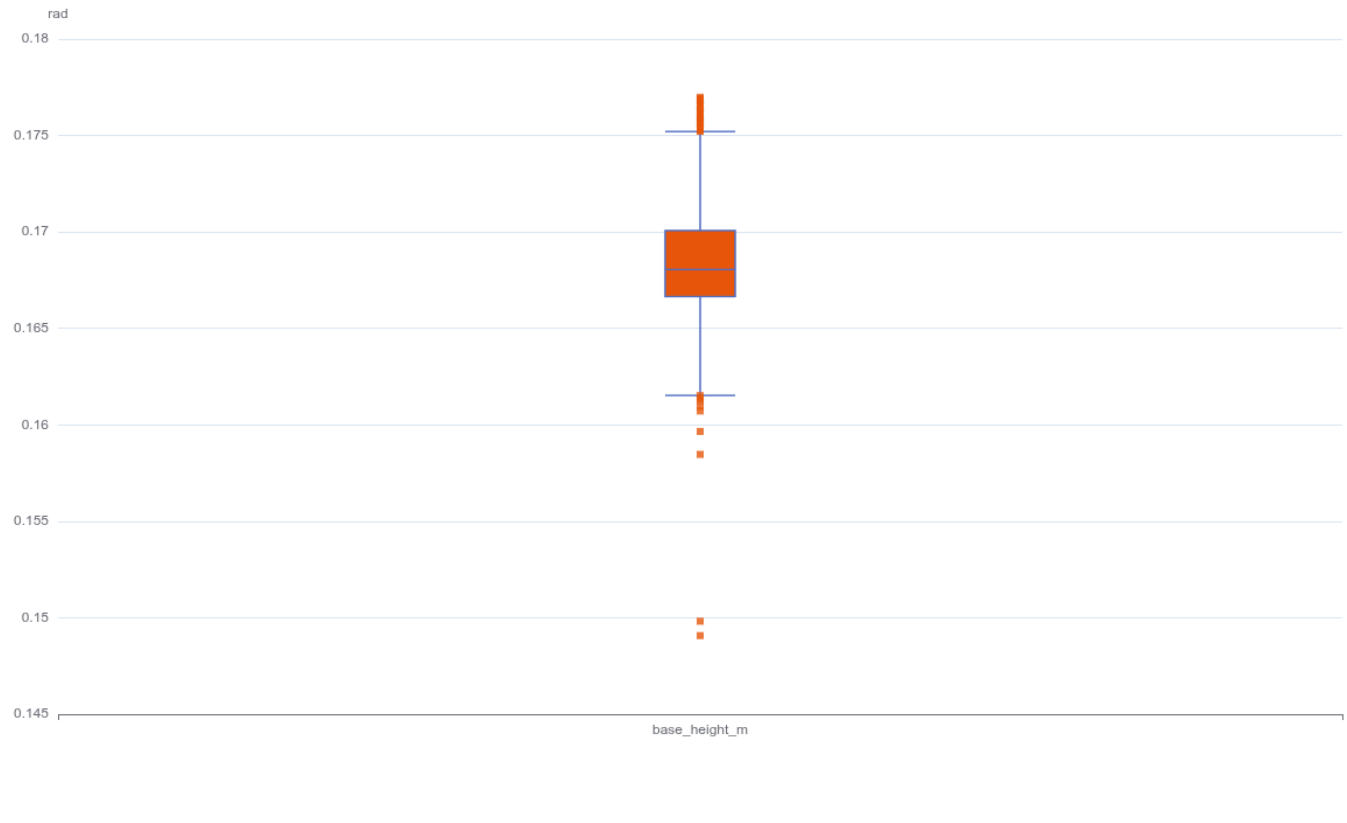

ratio
0.8
0.6
0.4
0.2
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

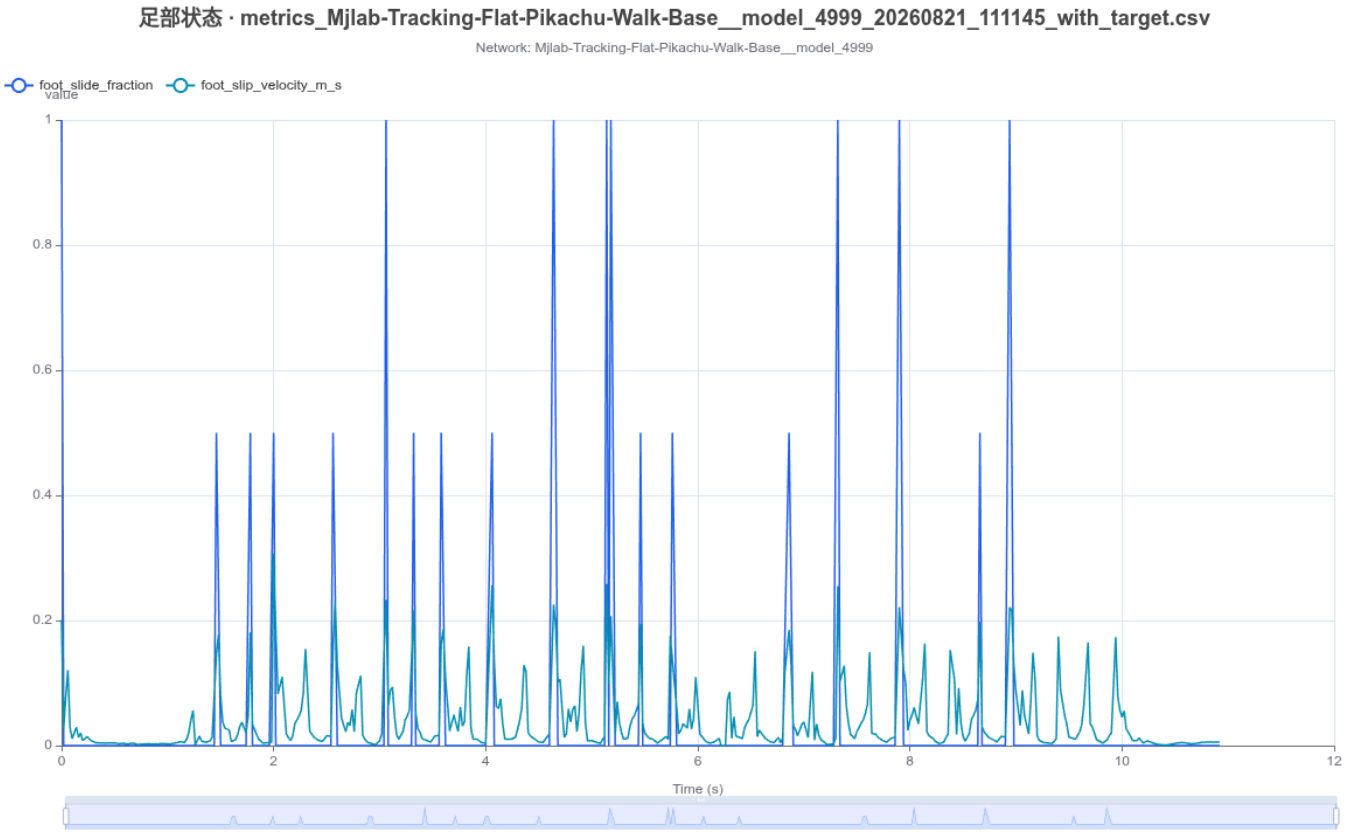
机体状态



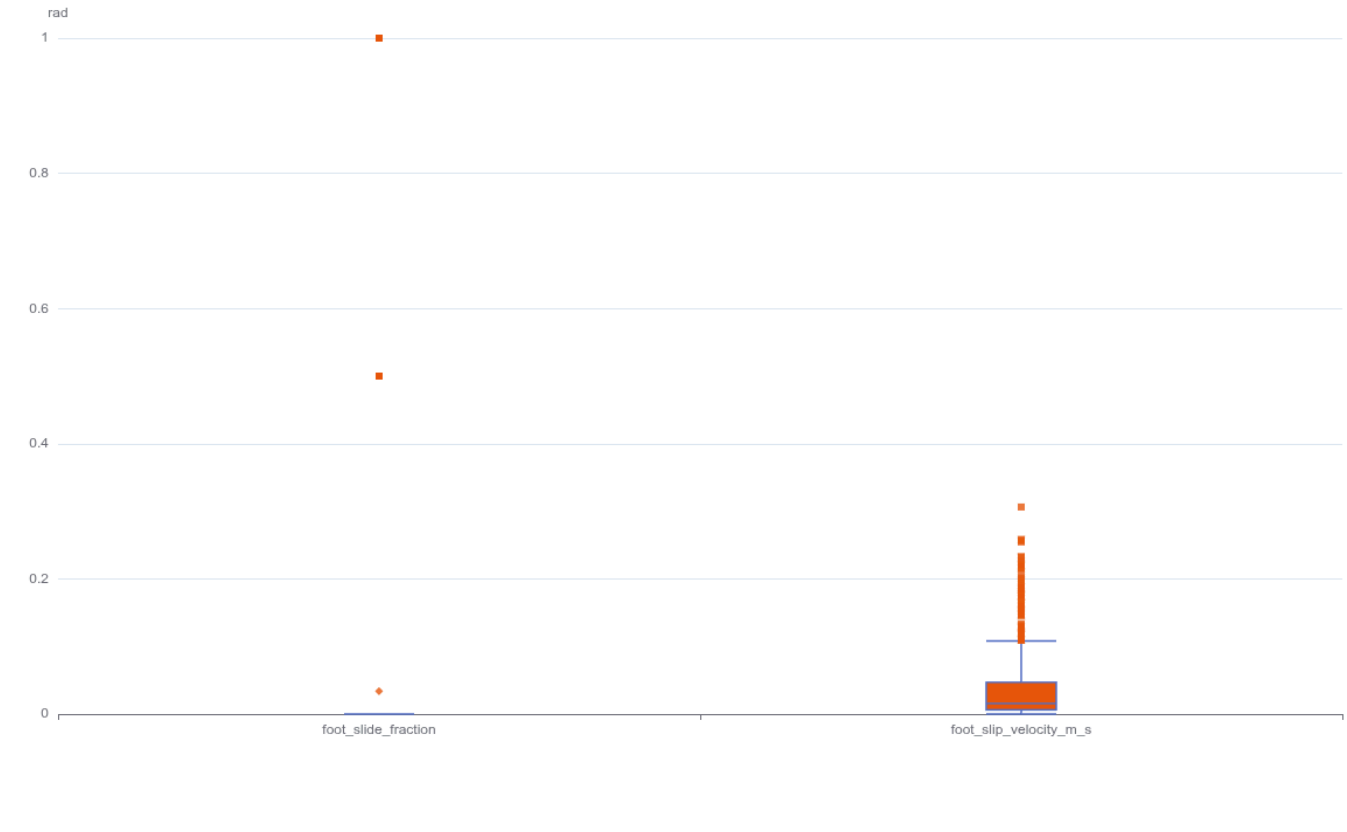
Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	503	0.149073	0.176984	0.168411	0.168449	0.176984	0.175641



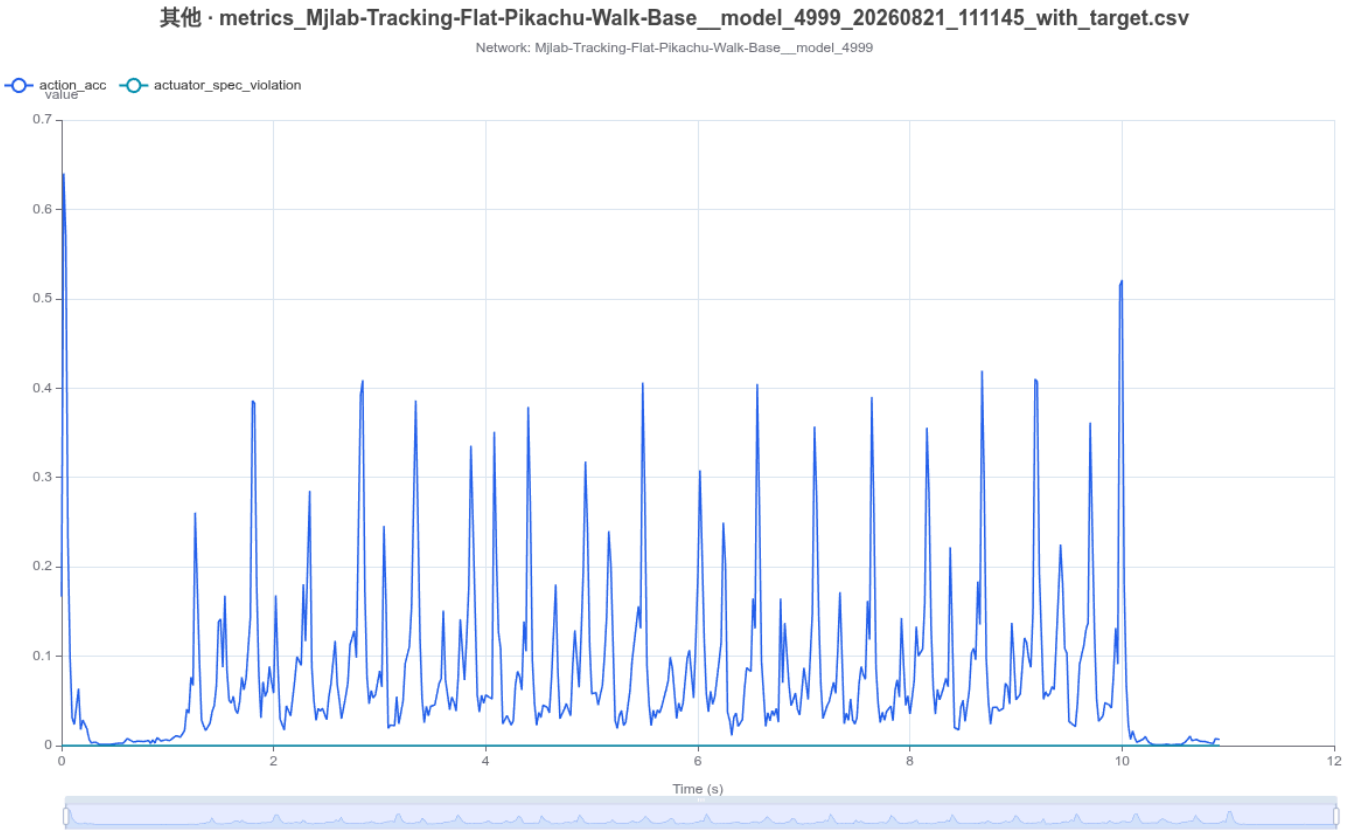
足部状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
foot_slide_fraction	–	–	503	0.00000	1.00000	0.0337972	0.157642	1.00000	0.00000
foot_slip_velocity_m_s	–	–	503	0.00000	0.306354	0.0392475	0.0666663	0.306354	0.164774



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	503	0.000589569	0.640203	0.0834788	0.125698	0.640203	0.280458
actuator_spec_violation	–	–	503	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

